

Financial ML Track

Final Project Report

Project Workspace · May 13, 2026

OBJECTIVE	Predict next-day direction of NIFTY 50 (binary: Up / Down) using OHLCV and derived features with walk-forward validation.
OUTCOME	Baseline logistic pipeline: accuracy ~0.476, AUC ~0.495 (near-random). Buy-and-hold outperformed strategy. Threshold tuning reduced losses but did not beat benchmark.
STATUS	Pipeline delivered. Model not deployable — further feature & model iteration required.

1 Problem Framing

The model targets **next-day closing direction** for the NIFTY 50 index using features computable at market close. The prediction horizon is one trading day, and the binary label is defined as **1** when the next-day close exceeds today's close, and **0** otherwise.

Baselines

Majority Class Accuracy	53.0%	Random Baseline (50/50)	50.0%
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Trade Policy & Loss Function

The ML model outputs probabilities fed into a **long/flat strategy**: enter long when predicted probability exceeds the threshold; otherwise stay flat. Transaction costs are modelled at **10 bps per position change**.

Why This Setup

- Daily direction prediction is a well-established micro-horizon benchmark.
- Cross-asset signals (Bank Nifty, India VIX) and technical features may provide modest predictive signal.
- Walk-forward validation eliminates lookahead leakage and simulates a realistic monthly refit cadence.

2 Data Audit

Coverage & Sources

Field	Detail
Date Range	2022-01-03 to 2025-12-31 (988 trading days)
NIFTY 50	nifty50.csv — OHLCV, clean. Minor volume zeros on a few rows.
Bank Nifty	banknifty.csv — cross-asset signal source.
India VIX	indiavix.csv — expected all-zero volume; 1 extreme spike observed.
Starter Features	starter_features.csv — 32 columns initially provided.

Feature Pruning — 5 Dropped Features

Feature	Reason for Removal
close_vs_252d_high	252-day lookback incompatible with walk-forward train windows
close_vs_252d_low	Same — 25.4% missingness
volume_ratio_20d	14.3% missingness — high imputation risk
dow	Calendar feature — overfitting risk
ma5_smooth_signal	Undocumented processing — leakage uncertain

Final Dataset

Total Rows	988	Features (after pruning)	26
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Up Days (label=1)	524 (53.0%)	Down Days (label=0)	464 (47.0%)
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3 Feature Engineering & EDA

After pruning and forward-fill / back-fill imputation, the final feature matrix contains **988 rows × 26 columns** with no missing values.

Group	Features
Returns (6)	ret_1d · ret_5d · ret_10d · ret_20d · ret_intraday · ret_overnight
Technical (10)	high_low_range · log_volume · close_vs_ma5 · close_vs_ma20 · close_vs_ma50 · momentum_5_20 · vol_5d · vol_20d · vol_50d · rsi_14
Cross-Asset (4)	bn_ret_1d · bn_ret_5d · nifty_bn_spread · nifty_bn_corr_20d
Vol Regime (4)	vix_level · vix_change · vix_5d_change · vix_ma_ratio
Statistical (2)	ret_zscore · volume_normalized

Key EDA Observations

- Moving-average related features are mutually correlated (expected by construction).
- ret_1d correlates with the ret_intraday + ret_overnight decomposition by identity.
- After ffill().bfill(), zero missing values remain in the final dataset.
- Numeric features are standardised inside the model pipeline (StandardScaler) — no data leakage from scaling.

4 Model & Training

Baseline Pipeline

StandardScaler → LogisticRegression (max_iter=1000, random_state=42)

Walk-Forward Scheme

Train Window	4 months (~80–85 days)	Test Window	1 month
Refit Cadence (Step)	1 month	Total Folds Generated	43

Splits are generated strictly by date (no shuffling). Training windows always precede test windows with zero overlap. The 4m/1m configuration provides a substantial training set while yielding many folds across the four-year dataset.

MLflow Tracking

- Experiment name: **nifty50_walkforward_training**
- Each fold runs as a nested MLflow run logging fold-level metrics and a saved model artifact.

5 Results

5a Baseline — Logistic Regression

Predictive Metrics (43-fold walk-forward)

Accuracy	0.4746 [CI: 0.441–0.505]	Balanced Accuracy	0.4672 [CI: 0.435–0.499]
ROC AUC	0.495 [CI: 0.458–0.532]	Majority Baseline	0.530

Backtest — Long/Flat, Threshold 0.50, TC 10 bps

Strategy Total Return	−17.4%	Benchmark (Buy & Hold)	+55.8%
Sharpe Ratio	−0.48	Max Drawdown	−19.96%
Win Rate	28.97%	Best Threshold (tuned)	0.30
Return at Best Threshold	≈ −3.55%	Sharpe at Best Threshold	≈ −0.04

Conclusion: Baseline model is near-random (AUC ~0.495) and not a deployable edge. Threshold tuning reduces losses but does not overcome the underlying weak signal.

5b Improved Model — LightGBM + Feature Selection + Hyperparameter Tuning

- Feature selection: top 10 features by LightGBM importance (computed on training set only).
- Tuning: RandomizedSearchCV (n_iter=10, TimeSeriesSplit n_splits=3) inside each walk-forward fold.

Predictive Metrics

Accuracy (mean)	0.503	Balanced Accuracy (mean)	0.492
ROC AUC (mean)	0.474	Folds	43

Backtest — Long-Short, Best Threshold 0.35, TC 10 bps

Strategy Total Return	−20.58%	Benchmark (Buy & Hold)	+55.8%
Sharpe Ratio	−0.447	Max Drawdown	−29.54%
Win Rate	49.61%		

Interpretation: LightGBM with feature selection and a long-short policy yields marginally worse backtest performance than the logistic baseline (−20.58% vs −17.4%). Long-short positions did not recover additional edge. This confirms that the underlying signal is weak or the feature set / horizon is insufficient for a profitable strategy.

6 Is the Edge Real? Statistical Significance & Robustness

Statistical Significance

- Bootstrapped 95% CIs for both accuracy and AUC include no clear separation from random chance.
- AUC confidence interval crosses 0.50 — no statistically significant predictive edge for the logistic baseline.

Walk-Forward Integrity

- Splits generated strictly by date; no shuffling applied at any stage.
- Training windows always precede test windows — no temporal leakage by construction.
- Features requiring long lookbacks (252-day) were removed to prevent silent leakage.

Random-Label Baseline

- Label-permutation within test windows yields AUC and accuracy indistinguishable from the fitted model.
- This is a strong negative signal: the model learns nothing beyond noise.

Backtest Caveats & Leakage Risks

- Residual leakage risk: any incorrectly computed feature could silently bias results.
- Execution assumptions: close-to-next-close fills assume perfect fills; real slippage is not captured beyond the 10 bps TC model.
- Threshold sweep shows small improvements at lower thresholds but no regime consistently outperforms buy-and-hold.

Recommended Next Experiments

#	Experiment	Rationale
1	Feature reduction to top 8–12 (importance / RFE on train only)	Reduce overfitting; sharpen signal-to-noise
2	LightGBM / XGBoost with constrained depth + early stopping	Non-linear learner; regularised to avoid overfit
3	Long-short strategy (short when $p < 1$ minus threshold)	Exploit symmetric information currently ignored
4	Alternative targets: $\pm 0.2\%$ threshold or multi-day horizon	Reduce label noise from tiny moves
5	Robustness: label-shuffle tests, 6m/1m folds, varied refit cadence	Stress-test pipeline assumptions

7 Conclusion & Deliverables

Summary

The data pipeline, walk-forward validation framework, MLflow experiment tracking, and a backtest engine have all been successfully delivered. The logistic baseline is **not a deployable edge**: predictive metrics are near-random and the backtest underperforms buy-and-hold by a wide margin. The improved LightGBM model does not materially change this conclusion.

The recommended path forward is to build a disciplined **feature selection and hyperparameter tuning pipeline fully inside the walk-forward loop** before claiming any predictive edge. Alternative targets (larger move thresholds, multi-day horizons) and a long-short policy should be explored simultaneously.

Artefact Index

Artefact	Path / Location
Feature matrix	data/processed/X_features.csv
Target vector	data/processed/y_target.csv
Date index	data/processed/dates.csv
Walk-forward metrics	data/processed/walkforward_metrics.csv
Walk-forward predictions	data/processed/walkforward_predictions.csv
Backtest time-series	data/processed/walkforward_backtest.csv
Backtest summary	data/processed/walkforward_backtest_summary.json
MLflow experiment	nifty50_walkforward_training (local artifact store)

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